Did PSD2 increase the adoption of bank-based payments relative to card payments in Europe?

Master Thesis in Economics - EC9901

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Abstract

This thesis evaluates whether the EU's Second Payment Services Directive (PSD2), implemented in 2018, shifted payment behavior toward bank-based account-to-account (A2A) payments relative to card payments. Using harmonized annual payment statistics from the ECB (EU countries) and the BIS Red Book (non-EU controls), I construct a country–year panel covering 2012–2023 and estimate a difference-in-differences model comparing EU member states to the United States and Canada, with Switzerland included in robustness checks. The main outcomes measure the share and relative intensity of credit transfers relative to card payments, primarily using transaction counts and complemented by value-based specifications. The baseline results points to a negative association between PSD2 implementation and the relative use of bank-based payments in EU countries after 2018. Although the estimates are economically meaningful in magnitude, they are not statistically significant under wild cluster bootstrap inference. The direction and magnitude of the coefficients remain stable across specifications, control sets, alternative outcome definitions, and expanded control groups. Event-study diagnostics reveal no clear structural break following implementation and suggest that post-2018 dynamics do not reflect a sustained shift toward bank-based payments. Overall, the findings provide no robust evidence that PSD2 generated a measurable reallocation from cards toward bank-based payments at the aggregate country level during the medium-run period studied (2018–2023). Rather than strengthening the competitive position of account-to-account payments in observable aggregates, PSD2 coincided with a continued relative prominence of card usage. These results contribute to the policy debate on open banking. If PSD2 had materially increased bank-based payment adoption, it would have provided strong empirical support for open banking as a competition-enhancing model. Instead, the evidence highlights a gap between regulatory ambition and observable market outcomes. This suggests that Europe's forthcoming PSD3 directive may need to focus less on expanding regulatory scope and more on ensuring that existing provisions translate into tangible consumer adoption and competitive effects.

1 Introduction

Electronic payments represent the backbone of modern economies. They determine how consumers transact, how firms bear transaction costs, and how competition unfolds among financial intermediaries. In Europe, card networks such as Visa and Mastercard have long dominated the retail payment landscape, raising concerns about market concentration, merchant fees, and limited consumer choice. To address these issues, policymakers sought to foster competition and innovation in retail payments through regulatory reform. The European Union's Second Payment Services Directive (PSD2) European Parliament and Council, 2015a was adopted in 2015 and implemented from January 2018, representing a landmark regulatory intervention in this area. Unlike its predecessor, PSD1 (European Parliament and Council, 2007), which primarily harmonised payment rules, PSD2 explicitly mandated open banking by requiring banks to grant licensed third-party providers access to payment accounts through the so-called access-to-account (XS2A) provision. It also introduced Strong Customer Authentication (SCA) to enhance security and reduce fraud. Together, these measures aimed to lower entry barriers, stimulate fintech competition, and rebalance market power between card-based networks and bank-based account-to-account (A2A) payment systems. This paper examines whether PSD2 achieved one of its core economic objectives: *Did PSD2 increase the adoption of bank-based payments relative to card payments in Europe?* If the directive was effective in shifting competitive dynamics, one would expect to observe a post-2018 increase in the share of credit transfers and other bank-based instruments relative to card transactions. Conversely, limited or absent effects would suggest that regulatory access alone may be insufficient to alter entrenched market structures. The empirical analysis employs a cross-country difference-in-differences design, comparing EU member states before and after 2018 to a control group of advanced non-EU economies. The results do not provide evidence of a statistically significant shift toward bank-based payments following PSD2 implementation. If anything, the estimated coefficients point toward a relative decline in the use of bank-based instruments compared to cards. Although estimates are imprecise, several economic mechanisms are consistent with this pattern. Payment habits may be highly persistent. Switching costs, including user familiarity, merchant acceptance infrastructure, and reward programs might limit rapid substitution away from established card networks. Even if PSD2 improved the technical feasibility of open banking-based payments, this may not have been sufficient to overcome the two-sided market advantages and scale economies embedded in global card networks. Moreover, implementation frictions, uneven API quality, and gradual consumer adoption may have delayed measurable aggregate effects. The contribution of this study is twofold. First, it provides a systematic, data-driven evaluation of a major EU financial regulation that

has thus far been assessed primarily through industry reports and qualitative policy analysis. Second, by linking PSD2's regulatory design to measurable changes in payment composition, the study contributes to the ongoing policy debate surrounding PSD3 (European Commission, 2023) and the future of open banking in Europe. If PSD2 had meaningfully shifted usage toward bank-based payments, this would have strengthened the case for open banking as a model for digital competition policy. However, the absence of such evidence instead highlights a potential gap between regulatory ambition and observable market outcomes. This suggests that future reforms may need to focus not only on expanding regulatory mandates but also on ensuring effective implementation, interoperability, and tangible consumer incentives. Understanding whether PSD2 altered payment behaviour is therefore both academically significant and directly relevant for European competition policy and digital financial infrastructure.

2 Institutional Background

The Payment Services Directives form the EU's main regulatory framework for retail payments. PSD1, adopted in 2007, aimed to create a single European market for payments by harmonising national regulations governing payment institutions, consumer rights, and cross-border transactions under one coherent legal framework. The objective was to reduce regulatory fragmentation across member states and facilitate cross-border provision of payment services. While PSD1 improved the efficiency of cross-border payments, it largely left the dominance of international card networks unchanged and did not create mechanisms for direct competition from bank-based alternatives European Parliament and Council (2007). PSD2, formally adopted in November 2015 (Directive (EU) 2015/2366), marked a substantial policy shift European Parliament and Council (2015a). Unlike PSD1, PSD2 introduced the principle of open banking by requiring banks to provide secure and standardised access to payment accounts for licensed third-party providers (TPPs). This access-to-account (XS2A) rule enabled fintech firms to initiate account-to-account (A2A) payments directly from consumers' bank accounts, subject to consumer consent. The directive was transposed into national law by January 13, 2018, at which point banks and payment providers became legally obliged to comply with the new access requirements. In parallel, concerns about card-market power and merchant fees were addressed through the Interchange Fee Regulation (Regulation (EU) 2015/751), which capped interchange fees for consumer card transactions European Parliament and Council (2015b). The implementation of PSD2 also included the introduction of Strong Customer Authentication (SCA), requiring two-factor authentication for most electronic payments European Banking Authority (2019). While SCA was intended to improve security and consumer trust, its rollout

generated transitional frictions, as enforcement varied across countries and was delayed in several cases. The stated objectives of PSD2 were multifaceted. The directive sought to (i) enhance security and reduce fraud through Strong Customer Authentication (SCA), (ii) encourage competition and innovation by mandating access to payment accounts for licensed third-party providers under the open banking framework, (iii) strengthen consumer protection and rights in electronic payments, and (iv) further integrate the European payments market by harmonising rules across member states (European Parliament and Council, 2015a; European Commission, 2022). Among these objectives, the competition and innovation dimension is most directly related to potential shifts in payment composition. By lowering barriers to entry and mandating account access through standardized interfaces, PSD2 aimed to enable bank-based account-to-account payment solutions to compete more directly with established card networks, particularly in online and point-of-sale contexts. Evaluating whether PSD2 measurably altered the relative use of bank-based versus card-based payments therefore provides an observable aggregate test of one of the directive's central economic ambitions. The policy relevance of PSD2 remains high, as policymakers are currently preparing PSD3, a follow-up directive intended to address perceived shortcomings and implementation inconsistencies under PSD2 (European Commission, 2023). Understanding whether PSD2 generated measurable shifts in payment composition is therefore directly relevant for the design and calibration of future EU payments regulation.

3 Literature Review

An increasing empirical literature has examined how PSD2 and the introduction of open banking have reshaped European payment markets. A key reference is Polasik (2020), who uses cross-country panel data from EU member states to analyse how PSD2 influenced the adoption of different payment instruments. By distinguishing between card payments and bank-based payments (primarily credit transfers), Polasik documents measurable changes in relative instrument shares. However, the analysis does not employ an explicit difference-in-differences framework comparing EU to non-EU countries, limiting causal interpretation of the observed trends. Additional institutional analyses by the European Central Bank (2023) and the International Monetary Fund (2024) examine open banking from a broader systemic perspective, focusing on financial stability, competition, and consumer protection (European Central Bank, 2023a; International Monetary Fund, 2024). While these reports highlight the transformative potential of open banking frameworks, they do not provide quasi-experimental evidence on whether PSD2 altered aggregate payment composition. At the micro level, Polasik and Kotkowski (2022) analyse consumer attitudes toward open banking services across 22 EU countries, emphasizing trust,

awareness, and perceived functionality as key drivers of adoption. Similarly, Briones de Araluze and Cassinello Plaza (2023) find that social influence and initial trust significantly shape consumers' willingness to adopt open-banking-based services. These studies suggest that regulatory access provisions alone may be insufficient to generate widespread behavioural change, as consumer adoption depends on perceptions, habits, and network effects. More broadly, the economics of payment systems emphasizes the role of two-sided markets, network externalities, and path dependence in sustaining incumbent dominance. Card networks operate as two-sided platforms characterized by network externalities and scale economies (Rochet and Tirole, 2003). Empirical evidence shows that card adoption benefits from reinforcing merchant and consumer participation (Rysman, 2007), thus, rapid substitution toward bankbased rails, even when regulatory barriers to entry are lowered, might be limited. In this context, the effectiveness of open banking as a competition-enhancing policy is ultimately an empirical question. Against this backdrop, there remains limited causal evidence on whether PSD2 measurably shifted aggregate payment composition toward bank-based account-to-account transfers relative to cards. This thesis contributes by implementing a cross-country difference-in-differences design comparing EU member states to advanced non-EU economies over 2012–2023. By focusing on relative payment shares and employing small-cluster robust inference, the analysis provides disciplined empirical evidence on whether PSD2 coincided with a detectable structural break in payment composition at the country level.

4 Data

4.1 Panel, Units, and Sample

The empirical analysis uses a country–year panel covering the period 2012–2023, subject to data availability and harmonisation constraints across sources. Each observation corresponds to country c in calendar year t. An annual frequency ensures comparability between EU member states and the non-EU control countries while aligning naturally with macroeconomic controls. The treated units are EU member states. In practice, the EU sample is restricted to countries with complete baseline coverage for the core payment series and harmonized availability of the macro controls used in the controlled specifications. Denmark (DK), Luxembourg (LU), and Malta (MT) are therefore excluded from the estimation sample. To ensure consistency and comparability across specifications, all baseline and robustness regressions, both with and without controls are estimated on this same restricted EU sample of 24 countries. The baseline control group consists of the United States and Canada. As a robustness exercise, Switzerland is added to the control group in an expanded specification. Countries that implemented open-banking mandates

comparable to PSD2 during 2018–2021 (e.g. Australia, Japan, Korea, Israel, Mexico, or the UK post-Brexit) are excluded from the control group to avoid contamination.

4.2 Core Sources

- **ECB Statistical Data Warehouse (SDW)**: Annual series for EU member states covering credit transfers, direct debits, and card payments.European Central Bank (2023b) Instant payments are reported as a subcategory of credit transfers and are therefore included within credit transfer aggregates.
- **BIS/CPMI Red Book**: Harmonised annual payment statistics for non-EU control countries (United States, Canada, and Switzerland).Bank for International Settlements, CPMI (2023)
- **World Bank**: GDP per capita (PPP, constant prices) and broadband penetration (subscriptions per 100 inhabitants).
- **European Commission / EBA**: Official PSD2 transposition dates and Strong Customer Authentication (SCA) implementation milestones.European Parliament and Council (2015a); European Banking Authority (2019)

4.3 Payment Instrument Definitions

For each country c and year t:

- CT_{ct}: Credit transfers (transaction counts in the baseline analysis; monetary values used in robustness checks). These represent account-to-account (A2A) push payments, including invoice transfers, checkout transfers, and peer-to-peer payments. Instant payments are included within this aggregate, since both ECB and BIS report them as a subset of credit transfers.
- DD_{ct}: Direct debits (counts and values). These are pull payments initiated by the payee (e.g. subscriptions and utility bills). Due to limited and discontinuous coverage in earlier years for several countries, direct debits are excluded from the baseline outcome construction and used only in post-2022 robustness analysis.
- Card_{ct}: Total card payments (counts and values), including debit and credit card transactions conducted at physical points of sale and online. ATM withdrawals are excluded, as they represent cash access rather than payment activity.

Credit transfers are the instrument most directly targeted by PSD2 through the access-to-account (XS2A) provisions. Card payments serve as the incumbent benchmark against which substitution is evaluated.

4.4 Outcome Variables

The analysis focuses on two relative measures constructed from transaction counts:

$$\text{BankShare}_{ct} = \frac{\text{CT}_{ct}}{\text{CT}_{ct} + \text{Card}_{ct}},$$
$$\text{BankCardRatio}_{ct} = \frac{\text{CT}_{ct}}{\text{Card}_{ct}}.$$

BankShare_{ct} measures the share of account-to-account payments relative to card payments, while $\text{BankCardRatio}_{ct}$ captures the relative intensity of credit transfer usage. Expressing outcomes as ratios controls for cross-country differences in payment system size and focuses the analysis on compositional shifts. For readability, I refer to these outcomes as *BankShare* and *BankCardRatio* in the text. Instant payments remain embedded within CT_{ct} to avoid double counting and ensure consistency across countries and time. Direct debits are excluded from baseline outcomes due to incomplete early coverage but are incorporated in a dedicated robustness exercise for the years 2022–2024. It is important to note that credit transfers constitute a broad aggregate that includes traditional bank transfers, invoice payments, peer-to-peer transfers, and instant payments. While PSD2 directly targeted account-to-account payment initiation services, the available data do not isolate PSD2-enabled transactions. The analysis therefore evaluates whether PSD2 altered the aggregate composition of credit transfers relative to cards, rather than measuring the direct uptake of specific open-banking payment products.

4.5 Control Variables

To account for macroeconomic and digital infrastructure trends, the regressions include:

- GDPpc_{ct}: GDP per capita (PPP, constant prices), capturing differences in economic development that may influence payment adoption.
- Broadband_{ct}: Broadband subscriptions per 100 inhabitants, proxying for digital infrastructure necessary for online and open-banking payments.

These variables vary across countries and over time and are included as X_{ct} in controlled specifications.

4.6 Construction and Harmonisation

All series are harmonised across ECB and BIS sources to ensure consistency in definitions and units of measurement. Naming conventions and instrument

classifications are aligned prior to constructing ratios. Missing observations are not imputed. Instead, regressions are estimated on the available balanced sample of countries with sufficient coverage in the baseline period. Direct debits are excluded from baseline outcome construction due to limited pre-2017 coverage, but incorporated in a post-2022 robustness specification. Any definitional breaks or coverage changes identified in the metadata are documented and interpreted cautiously.

4.7 Policy Timing Variables

The baseline difference-in-differences design defines:

$$\text{Post}_t = 1 \quad \text{for } t \geq 2018,$$

marking the PSD2 implementation period.

Event-time indicators are constructed as $k = \text{year} - 2018$ for the dynamic specification reported in the event-study analysis.

For descriptive purposes, countries are also classified according to baseline card exposure in 2015 (above- versus below-median), though this classification is not used in the main estimation and serves only to illustrate a conceptual heterogeneity extension.

Symbol	Name	Definition / Why it matters
CT_{ct}	Credit transfers	A2A push payments; includes instant and non-instant transfers under ECB definitions.
DD_{ct}	Direct debits	Pull payments; analysed in robustness and post-2022 composition checks due to limited early coverage.
Card_{ct}	Card payments	Debit/credit card transactions (excl. ATM); incumbent benchmark.
BankShare_ct	Bank share	$\frac{\text{CT}}{\text{CT} + \text{Card}}$; share of account-to-account payments relative to cards.
BankCardRatio_ct	Bank-to-card ratio	$\frac{\text{CT}}{\text{Card}}$; intensity of credit transfers relative to cards.
GDPpc_{ct}	GDP per capita	Dev. level; correlates with adoption of digital payments.
Broadband_{ct}	Broadband penetration	Digital infrastructure; prerequisite for online/open-banking use.

Table 1: Variables, definitions, and rationale (country–year panel, 2012–2023).

5 Empirical Strategy

5.1 Conceptual Mechanism

PSD2 reduced entry barriers in retail payments by requiring banks to provide secure, standardised access to customer accounts for licensed third-party providers. This "open banking" framework allowed third parties to initiate account-to-account (A2A)

payments directly from bank accounts, creating a channel that can compete with traditional card payments. If PSD2 materially shifted retail payment behaviour at the aggregate level, one would expect the relative use of credit transfers to increase compared to card payments after implementation.

5.2 Baseline Design: Cross-country Difference-in-Differences

The baseline design applies a difference-in-differences (DiD) framework comparing EU countries (treated) with non-EU control countries (United States and Canada) before and after PSD2 implementation in 2018. The use of the United States and Canada as controls is motivated by their advanced payments markets and stable institutional settings outside the EU regulatory regime during the sample window. Switzerland is added in a separate robustness branch to assess sensitivity to control-group composition. The estimand in this framework is the average differential change in aggregate payment composition in EU countries relative to non-EU controls after 2018. The analysis therefore captures whether PSD2 coincided with a measurable shift in country-level payment shares, rather than identifying the effect of specific PSD2-enabled technologies at the transaction level. This distinction is important when interpreting null results, given that the design evaluates observable aggregate substitution, not micro-level adoption dynamics.

- **Unit:** country–year panel.
- **Treatment:** EU member states subject to PSD2.
- **Controls:** United States and Canada (baseline); Switzerland added in robustness checks.
- **Outcomes:** share of credit transfers in (credit transfers + card payments) and the credit-transfer-to-card ratio.
- **Data:** ECB Statistical Data Warehouse (EU) and BIS/CPMI Red Book (non-EU); World Bank controls for GDP per capita and broadband penetration.

The baseline specification is:

$$y_{ct} = \alpha_c + \lambda_t + \beta\big(\mathrm{EU}_c \times \mathrm{Post}_t\big) + \gamma' X_{ct} + \varepsilon_{ct},$$

where y_{ct} denotes either $\mathrm{BankShare}_{ct}$ or $\mathrm{BankCardRatio}_{ct}$, α_c are country fixed effects, and λ_t are year fixed effects. The coefficient β captures the average post-2018 differential change in the EU relative to the control group. The control vector X_{ct} includes log GDP per capita and log broadband subscriptions (added sequentially in controlled specifications). Because the analysis uses a small number of country

clusters, inference relies on wild cluster bootstrap p-values in addition to clustered standard errors, following standard practice for DiD settings with few clusters. Cameron et al. (2008a)

5.3 Dynamic diagnostic: Event-study specification

To assess the plausibility of the parallel-trends assumption beyond visual pre-trend plots, I estimate an event-study specification that replaces the single post indicator with relative-time indicators around 2018. Event time is defined as $k = \text{year} - 2018$, with $k = -1$ (2017) as the omitted reference period. The specification includes country and year fixed effects and traces the evolution of EU outcomes relative to the control group before and after PSD2 implementation. The event-study is interpreted as a diagnostic complement to the baseline DiD estimates.

5.4 Identification Assumptions and Checks

The key identifying assumption is that, absent PSD2, EU and control countries would have followed similar outcome trajectories (parallel trends assumption). I evaluate this assumption in several ways. I begin by presenting descriptive pre-trend figures for 2012–2017 comparing EU and control group means in levels and indexed form. I then estimate the event-study specifications to examine whether pre-treatment coefficients (leads) are close to zero relative to the reference year. As an additional falsification exercise, I implement a placebo difference-in-differences test by assigning a fake treatment year prior to 2018. If the baseline results were driven by spurious structural differences or mechanical divergence between EU and control countries, one would expect to observe a similarly strong effect under an arbitrary reform date. The placebo test therefore serves as a diagnostic check against false-positive identification driven by pre-existing trends rather than by PSD2 implementation. To mitigate omitted-variable concerns, I add a small set of time-varying controls capturing economic development and digital infrastructure (GDP per capita and broadband penetration). These are included sequentially to evaluate coefficient stability. Finally, I assess measurement robustness by re-estimating the baseline models using transaction values instead of counts, and by checking sensitivity to the definition of "bank-based payments" when direct debits are included in a restricted late-period sample where direct-debit coverage is adequate.

5.5 Measurement Robustness and Interpretation

The baseline analysis uses transaction *counts* as the preferred measure because counts capture frequency of use. However, as a robustness check, I replicate the outcomes using transaction *values* and compare results to the counts-based estimates.

Counts baseline:

$$\text{BankShare}_{ct}^{\text{cnt}} = \frac{\text{CT}_{ct}^{\text{cnt}}}{\text{CT}_{ct}^{\text{cnt}} + \text{Card}_{ct}^{\text{cnt}}},$$
$$\text{BankCardRatio}_{ct}^{\text{cnt}} = \frac{\text{CT}_{ct}^{\text{cnt}}}{\text{Card}_{ct}^{\text{cnt}}}.$$

Monetary values robustness:

$$\text{BankShare}_{ct}^{\text{val}} = \frac{\text{CT}_{ct}^{\text{val}}}{\text{CT}_{ct}^{\text{val}} + \text{Card}_{ct}^{\text{val}}},$$
$$\text{BankCardRatio}_{ct}^{\text{val}} = \frac{\text{CT}_{ct}^{\text{val}}}{\text{Card}_{ct}^{\text{val}}}.$$

Direct debits are excluded from the baseline outcomes because harmonised annual direct-debit series are short and discontinuous in earlier years. Instead, I use direct debits in a separate robustness exercise that redefines the bank-based share to include DD in a restricted 2022–2024 sample where coverage is sufficient, providing a check on sensitivity to the definition of "bank-based payments".

PSD2 → open-banking access → potentially greater competition → relative shift in payment mix

Interpreting the estimates, a relative increase in credit-transfer usage compared with cards would be consistent with PSD2 facilitating A2A substitution. Conversely, null or negative relative changes would suggest limited aggregate substitution toward bank-based payments over the period studied, potentially reflecting frictions such as uneven API quality, merchant adoption constraints, or persistent consumer and merchant preferences for card-based instruments.

6 Results

6.1 Descriptive evidence and pre-trends (2012–2017)

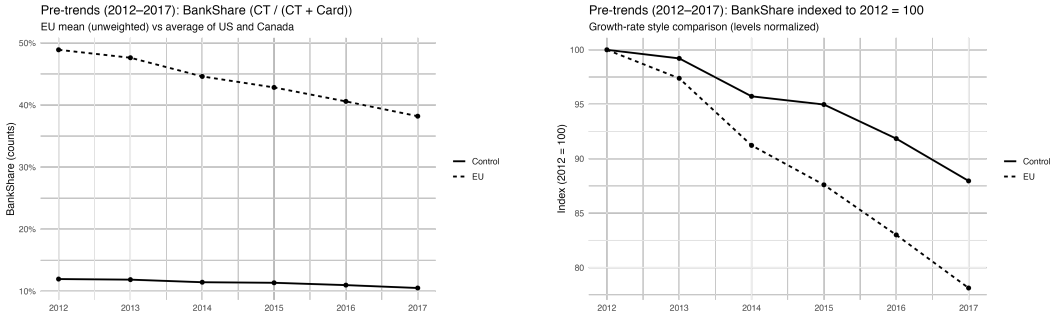
Using annual payment statistics harmonised across the ECB (EU member states) and the BIS (non-EU countries), I construct two baseline outcomes based on transaction counts: (i) the credit-transfer share of non-cash payments,

$$\text{BankShare}_{ct} = \frac{CT}{CT + Card},$$

and (ii) the ratio of credit transfers relative to card payments,

$$\text{BankCardRatio}_{ct} = \frac{CT}{Card}.$$

These measures capture the relative use of bank-based account-to-account (A2A) payments (proxied by credit transfers, CT) versus card payments ($Card$). Following the baseline design, I focus on counts (rather than values) and exclude direct debits from the main outcome construction due to limited and discontinuous coverage across countries. Before estimating a difference-in-differences (DiD) model, I assess the plausibility of the parallel trends assumption by comparing pre-treatment trends between the treated group (EU member states) and the control group (United States and Canada). The EU sample is restricted to countries with complete coverage of credit transfer and card payment counts in the baseline window, while the control group comprises BIS-reported series for the United States and Canada. Figures 1a–1b plot group means for BankShare_{ct} over 2012–2017 (pre-PSD2). The corresponding pre-trend plots for $\text{BankCardRatio}_{ct}$ are reported in the Appendix. Both outcomes display smooth and broadly similar downward trends, though not identical, with differences between the EU and control group being primarily in levels rather than slopes. Consistent with the level plots, the EU starts from a higher baseline level of bank-based payment usage than the control group throughout the pre-period. To assess relative changes more directly, the indexed versions normalise each group to 2012 = 100. The indexed trends are broadly aligned across groups for both outcomes, although the EU shows a somewhat steeper decline in parts of the pre-period. Taken together, the pre-treatment patterns provide broadly supportive, though not definitive visual evidence for the plausibility of parallel trends prior to PSD2 implementation. In addition, I report raw (unadjusted) group means over the full sample period 2012–2023 for both outcomes (Figures 2 and 3). These figures show that EU countries experienced a steady and persistent decline in the relative use of credit transfers well before 2018, while the control group remained comparatively flat at a lower baseline level. Importantly, there is no visible structural break in 2018 in the raw EU mean trajectories: the downward trend in the EU appears smooth and continuous across the reform year. That said, the absence of a visible break in 2018 suggests that PSD2 did not interrupt or reverse an already ongoing shift toward card-based payments in observable aggregate composition data. Note that these plots are descriptive and do not establish causality; they are used to assess whether pre-treatment dynamics are sufficiently similar to support the DiD identification strategy.



(a) Levels (2012–2017) **(b)** Indexed to 2012 = 100

Figure 1: Pre-treatment trends in $\text{BankShare}_{ct} = CT/(CT + Card)$, EU versus control group. EU mean (unweighted) compared with the average of the United States and Canada, 2012–2017.

Note: The treated group includes 24 EU countries (excluding Denmark, Luxembourg, and Malta due to missing control data), and the control group includes 2 countries (United States and Canada), for a total of 26 country clusters.

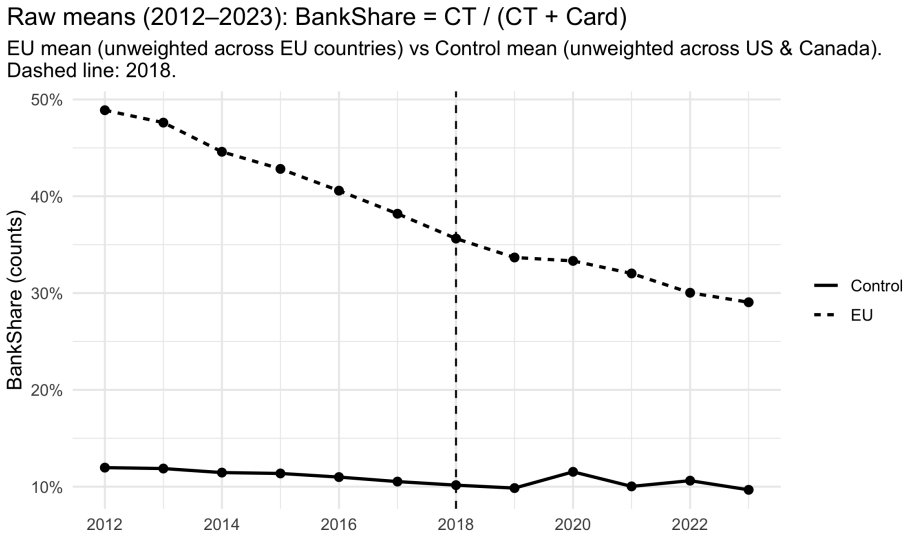


Figure 2: Raw group means (2012–2023): BankShare (counts). EU mean (unweighted) versus the average of the United States and Canada. The vertical line marks 2018.

*Note:*The treated group includes 24 EU countries (excluding Denmark, Luxembourg, and Malta due to missing control data), and the control group includes 2 countries (United States and Canada), for a total of 26 country clusters.

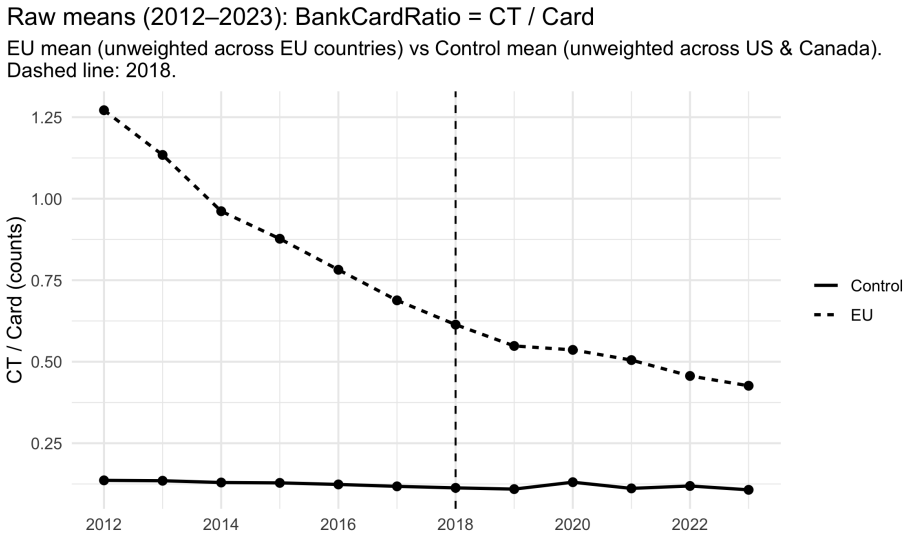


Figure 3: Raw group means (2012–2023): BankCardRatio (counts). EU mean (unweighted) versus the average of the United States and Canada. The vertical line marks 2018.
*Note:*The treated group includes 24 EU countries (excluding Denmark, Luxembourg, and Malta due to missing control data), and the control group includes 2 countries (United States and Canada), for a total of 26 country clusters.

6.2 EU country trajectories

To examine whether the aggregate null result masks heterogeneous country responses, I plot individual EU country trajectories for the BankShare outcome over 2012–2023 (Figure 13 in the Appendix). Thin lines represent individual countries, while the thick line shows the unweighted EU mean. The figure reveals substantial cross-country variation in levels, but a broadly shared downward trend over time. Importantly, there is no visually distinct structural break around 2018 in most countries. This pattern suggests that the aggregate results are not driven by a small number of extreme country outliers, but rather reflect a common evolution across member states.

6.3 Baseline Difference-in-Differences Results

Table 2 reports baseline difference-in-differences estimates of the association between the introduction of PSD2 and the relative use of bank-based account-to-account payments. The table compares EU countries to a control group consisting of the United States and Canada, using a post-2018 indicator (Post = 1 for years $\geq$ 2018) to capture the PSD2 implementation period. The analysis is restricted to a fixed complete-case EU sample (24 countries), excluding Denmark, Luxembourg, and Malta due to missing GDP per capita and broadband controls in 2012–2023; the same EU sample is used in both baseline and controlled specifications to ensure comparability. Panel A presents results for the share of credit transfers in total non-cash payments (BankShare), while Panel B reports estimates for the ratio of credit transfers to card payments (BankCardRatio). In both cases, the coefficient

of interest captures the interaction between EU membership and the post-2018 period. The point estimates are negative across both outcome measures, indicating a relative decline in the use of bank-based payments in EU countries following the introduction of PSD2 compared to the control group. Standard errors are clustered at the country level, and statistical inference is based on wild cluster bootstrap p-values to account for the limited number of clusters. With few clusters, conventional cluster-robust standard errors can substantially over-reject the null hypothesis and overstate precision (Cameron et al., 2008b; Cameron and Miller, 2015). Accordingly, wild cluster bootstrap p-values are treated as the primary benchmark for inference, as they provide more reliable finite-sample inference in settings with a limited number of clusters. While the point estimates are non-zero in magnitude, they are not statistically significant at conventional levels. These estimates are interpreted cautiously: they are causal under the parallel-trends assumption, which is assessed using pre-trend figures and event-study diagnostics, but cannot be verified directly. The purpose of Table 2 is to establish a transparent and disciplined starting point for the empirical analysis, grounded in descriptive pre-treatment trend diagnostics and a minimal specification. Subsequent sections examine the robustness of these findings to the inclusion of country-level controls, alternative specifications, and dynamic treatment effects.

Table 2: Baseline Difference-in-Differences Estimates: Bank-Based vs Card Payments

	Outcome	
	BankShare (counts)	BankCardRatio (counts)
EU × Post(2018+)	**-0.104**	**-0.426**
	(0.021)	(0.107)
Wild bootstrap p-value	0.124	0.141
Observations	309	309
Countries	26	26
Years	2012–2023	2012–2023

Note: This table reports baseline difference-in-differences estimates of the association between PSD2 implementation and the relative use of bank-based account-to-account payments. The treatment indicator equals one for EU countries and zero for the control group (United States and Canada). The post-treatment period begins in 2018. Outcomes are constructed using payment *counts* only: *BankShare* is defined as the ratio of credit transfer transactions to the sum of credit transfer and card transactions, and *BankCardRatio* is defined as the ratio of credit transfer transactions to card transactions. All specifications include country and year fixed effects. Standard errors are clustered at the country level, and statistical inference is based on wild cluster bootstrap p-values with 999 replications. The sample includes EU countries with complete baseline coverage and spans the years 2012–2023. The treated group includes 24 EU countries (excluding Denmark, Luxembourg, and Malta due to missing control data), and the control group includes 2 countries (United States and Canada), for a total of 26 country clusters.

6.4 Baseline results with controls

Table 3 extends the baseline difference-in-differences specification by sequentially adding macroeconomic and digital infrastructure controls. Across both outcome measures, the estimated effect of PSD2 remains negative and of similar magnitude to the baseline specification without controls. For the bank share of payments, the estimated coefficient declines modestly from -0.10 in the baseline model to -0.08 when controlling for GDP per capita and broadband penetration. A similar pattern is observed for the bank-to-card payment ratio, where the coefficient attenuates from -0.43 to -0.30 across specifications. In all cases, the direction of the estimated effect is unchanged. The stability of the estimates across specifications suggests that differential income growth or broadband expansion are unlikely to be the sole drivers of the baseline EU–control differences. While none of the estimates are statistically significant under wild cluster bootstrap inference, the consistency in sign and magnitude supports the interpretation that PSD2 yields consistently negative point estimates, a relative shift away from bank-based account-to-account payments toward card payments.

Table 3: Baseline difference-in-differences estimates with controls

	(1) No controls	(2) + log GDPpc	(3) + log GDPpc + log(1+Broadband)
Panel A. BankShare (counts)			
EU × Post(2018+)	**-0.104**	**-0.090**	**-0.082**
	(0.021)	(0.027)	(0.025)
Wild bootstrap p-value	0.124	0.146	0.157
Panel B. BankCardRatio (counts)			
EU × Post(2018+)	**-0.426**	**-0.361**	**-0.303**
	(0.107)	(0.110)	(0.101)
Wild bootstrap p-value	0.157	0.168	0.175
Observations	309	309	309
Countries	26	26	26
Years	2012–2023	2012–2023	2012–2023
Country FE	Yes	Yes	Yes
Year FE	Yes	Yes	Yes

Note: This table reports difference-in-differences estimates of the effect of PSD2 on payment composition using country–year panel data. The treatment indicator equals one for EU member states after 2018. Outcomes are constructed using payment counts only. All specifications include country and year fixed effects. Standard errors clustered at the country level are reported in parentheses. Wild cluster bootstrap p-values (B = 999) are reported beneath coefficient estimates. Control variables are added sequentially and include log GDP per capita (PPP, constant prices) and log broadband subscriptions per 100 inhabitants. The treated group includes 24 EU countries (excluding Denmark, Luxembourg, and Malta due to missing control data), and the control group includes 2 countries (United States and Canada), for a total of 26 country clusters.

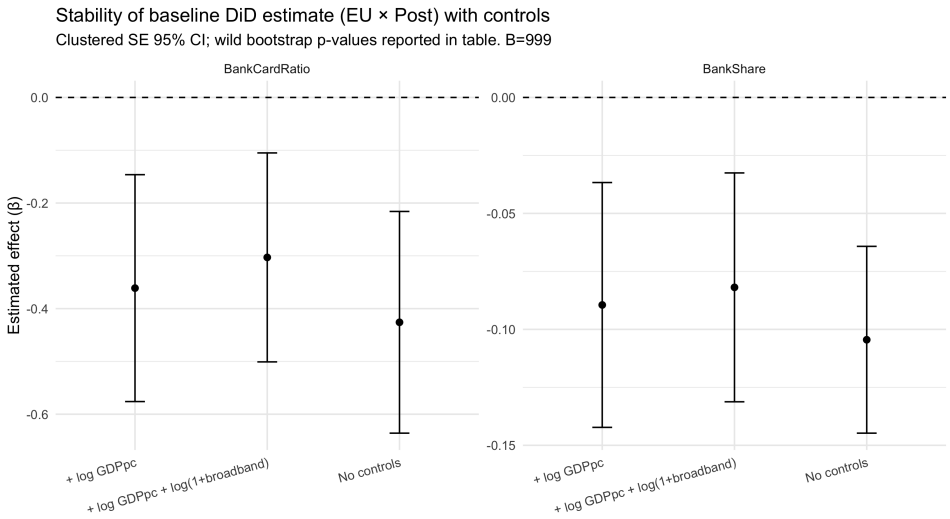


Figure 4: Stability of baseline DiD estimates with controls

Note: The figure plots point estimates and 95% confidence intervals for the EU × Post coefficient across specifications. Estimates are based on country–year panel regressions with country and year fixed effects. Confidence intervals use clustered standard errors at the country level. Wild cluster bootstrap p-values (B = 999) are reported in the corresponding regression table. The treated group includes 24 EU countries (excluding Denmark, Luxembourg, and Malta due to missing control data), and the control group includes 2 countries (United States and Canada), for a total of 26 country clusters.

6.5 Event-study diagnostic: BankShare (counts-based)

As a diagnostic for the difference-in-differences (DiD) identifying assumption, I estimate an event-study specification that replaces the single post indicator with a set of relative-time indicators around the PSD2 implementation year (2018). The coefficients trace the evolution of the treated group (EU) relative to the control group (US/Canada) over event time $k = \text{year} - 2018$, with $k = -1$ (2017) as the reference period. All specifications include country and year fixed effects, and inference is based on standard errors clustered at the country level. Figure 5 shows the event-study coefficients for the BankShare outcome (constructed from counts only). The post-2018 coefficients are predominantly negative, consistent with the baseline DiD estimates. The pre-treatment coefficients (leads) are generally close to zero near the reference year ($k = -1$), but exhibit some non-flatness earlier in the pre-period. Taken together with the indexed pre-trend plots, this pattern is best interpreted as supportive but not definitive evidence for strict parallel trends. Accordingly, the event-study is used as a diagnostic complement rather than a standalone causal estimate.

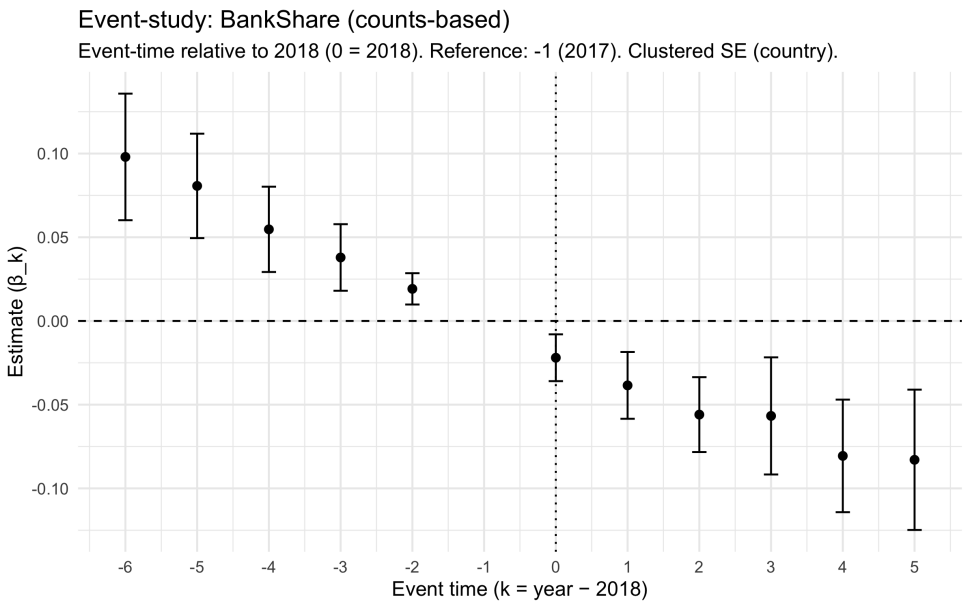


Figure 5: Event-study estimates for BankShare (counts-based). Coefficients are relative to 2017 ($k = -1$), with event time defined as $k = \text{year} - 2018$ (so $k = 0$ corresponds to 2018). Country and year fixed effects are included. Standard errors are clustered at the country level. *Note:*The treated group includes 24 EU countries (excluding Denmark, Luxembourg, and Malta due to missing control data), and the control group includes 2 countries (United States and Canada), for a total of 26 country clusters.

6.6 Placebo Difference-in-Differences Test

As an additional diagnostic of the identification strategy, I implement a placebo difference-in-differences test by assigning a fictional treatment year of 2016. Specifically, I redefine the post indicator as $\text{PostPlacebo}_t = 1$ for years $t \geq 2016$, while keeping the estimation sample, fixed effects structure, and inference procedure identical to the baseline specification. If the baseline estimates were driven by spurious structural differences between EU and control countries rather than PSD2 implementation itself, one would expect to observe a similarly strong effect under this placebo timing. Table 4 reports the placebo estimates for both outcomes using transaction counts. The placebo EU × Post coefficient is negative but statistically indistinguishable from zero under wild cluster bootstrap inference for both BankShare and BankCardRatio. Figure 6 compares the baseline and placebo point estimates. The absence of a statistically significant placebo effect suggests that the baseline results are not driven by an arbitrary structural break or mechanical divergence between EU and control countries tied to an alternative reform timing. The similarity in sign and magnitude between baseline and placebo coefficients is consistent with gradual underlying trends rather than a discrete reform-induced shift. Taken together with the event-study diagnostics, the placebo test reinforces the interpretation that PSD2 did not generate a detectable structural break in aggregate payment composition.

Table 4: Placebo difference-in-differences estimates (Post ≥ 2016)

	BankShare (counts)	BankCardRatio (counts)
EU × PostPlacebo(≥ 2016)	**-0.107**	**-0.478**
	(0.020)	(0.121)
Wild bootstrap p-value	0.121	0.142
Observations	309	309
Countries	26	26
Years	2012–2023	2012–2023

Note: The placebo specification assigns a fictitious treatment year of 2016, defining PostPlacebo = 1 for years ≥ 2016. The estimation sample, fixed effects structure, clustering, and wild cluster bootstrap inference procedure (B = 999) are identical to the baseline specification. The treated group includes 24 EU countries; the control group includes the United States and Canada (26 clusters in total).

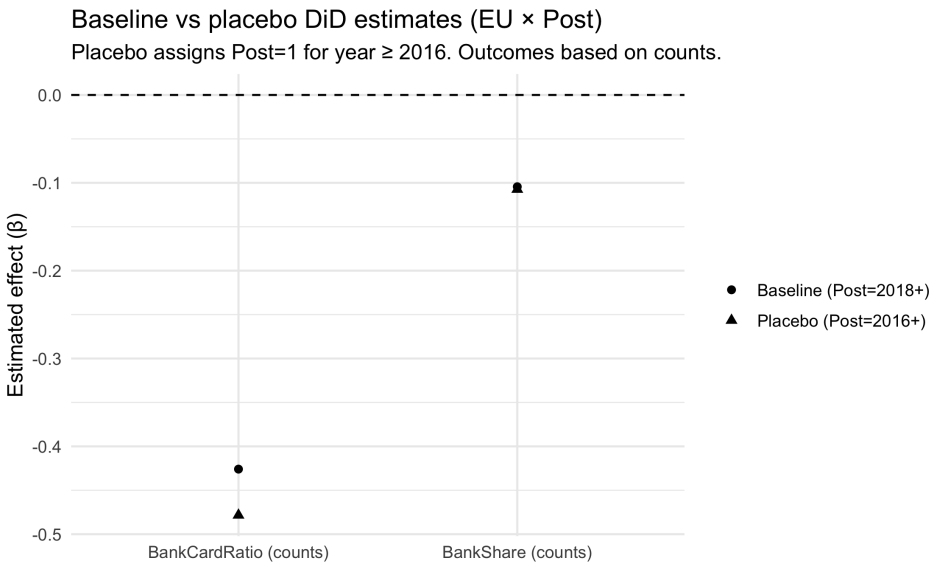


Figure 6: Baseline vs placebo DiD estimates (EU × Post). The placebo assigns Post = 1 for years ≥ 2016. Outcomes based on transaction counts.
*Note:*The treated group includes 24 EU countries (excluding Denmark, Luxembourg, and Malta due to missing control data), and the control group includes 2 countries (United States and Canada), for a total of 26 country clusters.

7 Robustness Analysis

7.1 Values-based estimates with controls

In this robustness check, I re-estimate the difference-in-differences specification using transaction *values* rather than counts as outcomes. Specifically, I consider (i) the share of credit transfer values relative to total non-cash payment values, and (ii) the ratio of credit transfer values to card payment values. The specification mirrors the baseline counts-based model in terms of fixed effects structure and inference strategy, including country and year fixed effects, country-clustered standard errors, and wild cluster bootstrap inference.

Table 5 reports estimates sequentially adding controls for economic development and digital infrastructure: (i) no controls, (ii) log GDP per capita, and (iii) log GDP per capita and log broadband penetration. The estimated treatment effect for the values-based BankShare outcome remains negative across all specifications. While coefficient magnitudes attenuate modestly after the inclusion of controls, there is no sign reversal or pronounced instability, reinforcing the directional consistency of the baseline counts-based findings. For completeness, the baseline values-based specification without controls is reported in Table 9 in the Appendix. Consistent with the controlled estimates, the point estimates are negative but statistically imprecise under wild cluster bootstrap inference. Given that payment values are more volatile than transaction counts and more sensitive to cross-country price levels, large-ticket transactions, and institutional reporting differences, these results are interpreted as complementary evidence rather than the primary estimand.

Table 5: Difference-in-differences estimates using values with controls

Spec	(1) No controls	(2) + log GDPpc	(3) + log GDPpc + log(1+Broadband)
EU × Post(2018+)	**-0.017**	**-0.020**	**-0.019**
	(0.007)	(0.007)	(0.007)
Wild bootstrap p-value	0.187	0.226	0.202
Observations	309	309	309
Countries	26	26	26
Years	2012–2023	2012–2023	2012–2023
Country FE	Yes	Yes	Yes
Year FE	Yes	Yes	Yes

*Note:*The treated group includes 24 EU countries (excluding Denmark, Luxembourg, and Malta due to missing control data), and the control group includes 2 countries (United States and Canada), for a total of 26 country clusters.

Figure 7 visualises the corresponding coefficient estimates with clustered 95% confidence intervals. The values-based point estimates for BankShare are negative, consistent with the counts-based results, but remain imprecisely estimated. This pattern suggests that the absence of positive effects is not driven by the choice of measurement unit, but reflects a broader stability in the relative use of bank-based versus card payments.

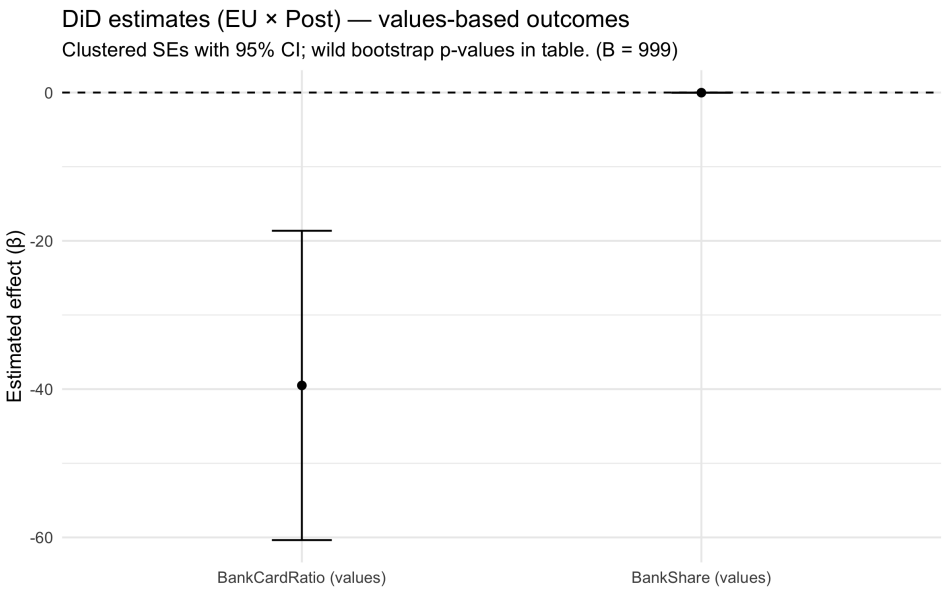


Figure 7: Values-based DiD estimates (EU × Post). Point estimates and 95% confidence intervals for the EU × Post coefficient using transaction values. Confidence intervals are based on country-clustered standard errors; wild cluster bootstrap p-values are reported in Table 9 (B = 999).
*Note:*The treated group includes 24 EU countries (excluding Denmark, Luxembourg, and Malta due to missing control data), and the control group includes 2 countries (United States and Canada), for a total of 26 country clusters.

Figure 8 illustrates the stability of the estimated EU × Post coefficient across control specifications. The point estimates remain consistently negative and exhibit only modest attenuation when controls are added, with overlapping confidence intervals across all models. This reinforces the conclusion that the values-based results are not sensitive to the inclusion of standard macroeconomic controls and that the absence of statistically significant effects does not reflect specification instability.

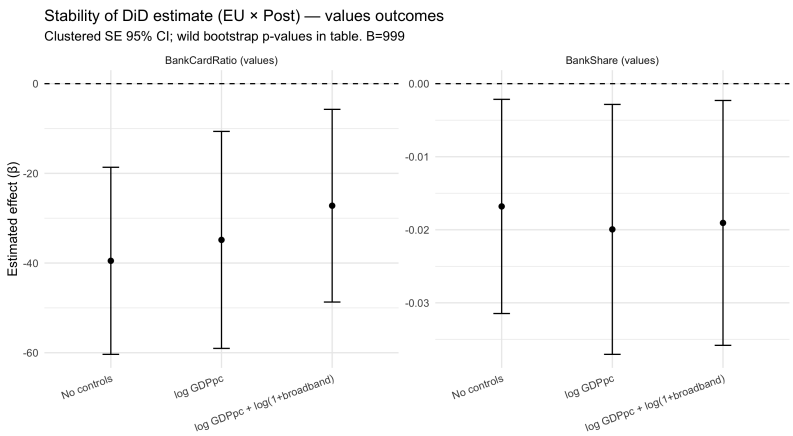


Figure 8: Stability of DiD estimates (EU × Post) for values-based outcomes across control specifications. Error bars indicate 95% confidence intervals based on country-clustered standard errors. Wild cluster bootstrap p-values are reported in the corresponding table.
*Note:*The treated group includes 24 EU countries (excluding Denmark, Luxembourg, and Malta due to missing control data), and the control group includes 2 countries (United States and Canada), for a total of 26 country clusters.

7.2 Robustness to Alternative Control Group: Switzerland

As an additional robustness exercise, I extend the non-EU control group to include Switzerland. Switzerland represents a high-income economy with an advanced electronic payments infrastructure that was not subject to PSD2, making it a natural candidate for an expanded control group. Due to reporting conventions in the BIS CPMI dataset, Swiss card payment statistics are reported for transactions executed at payment terminals inside the country, which constitutes the broadest card aggregate available for Switzerland. This restriction is time-invariant and therefore does not mechanically affect the difference-in-differences identification. Table 6 and Table 7 show that including Switzerland does not materially alter the estimated treatment effects. Point estimates remain negative and of similar magnitude to the baseline specification, and confidence intervals overlap with those obtained using the original control group. This indicates that the qualitative findings are not sensitive to modest expansions of the control group.

Table 6: Baseline difference-in-differences estimates (counts) with expanded control group including Switzerland

	BankShare (counts)	BankCardRatio (counts)
EU × Post(2018+)	**-0.080**	**-0.339**
	(0.029)	(0.130)
Wild bootstrap p-value	0.076	0.084
Observations	321	321
Countries	27	27
Years	2012–2023	2012–2023
Country FE	Yes	Yes
Year FE	Yes	Yes
Controls	No	No

Note: Wild cluster bootstrap p-values fall below 0.10 in the specification without controls, but remain above 0.05. This is best interpreted as *suggestive* evidence under the expanded control group rather than a statistically robust effect. The treated group includes 24 EU countries (excluding Denmark, Luxembourg, and Malta due to missing control data), and the control group includes 2 countries (United States and Canada), for a total of 26 country clusters.

Table 7: Table Baseline DiD estimates with controls, including Switzerland

	BankShare (counts)			BankCardRatio (counts)		
Spec	(1) No ctrls	(2) + log GDPpc	(3) + log GDPpc + log(1+BB)	(1) No ctrls	(2) + log GDPpc	(3) + log GDPpc + log(1+BB)
EU × Post(2018+)	**-0.080**	**-0.089**	**-0.082**	**-0.339**	**-0.361**	**-0.303**
	(0.029)	(0.027)	(0.025)	(0.130)	(0.110)	(0.101)
Wild bootstrap p-value	0.076	0.176	0.176	0.092	0.142	0.161

Interpretation: This robustness check addresses sensitivity to the choice of control countries; it does not by itself validate the parallel-trends assumption.

Note: Table 6 reports difference-in-differences estimates using the expanded control group that includes Switzerland in addition to the United States and Canada. The specification mirrors the baseline controlled models in the main text, sequentially adding log GDP per capita and broadband penetration. All regressions include country and year fixed effects, and inference is based on country-clustered standard errors with wild cluster bootstrap p-values.
Across all specifications and both outcome measures, the estimated EU × Post coefficient remains negative and of similar magnitude to the baseline results. The inclusion of Switzerland therefore does not materially alter the estimated association between PSD2 implementation and the relative use of bank-based payments. This robustness check suggests that the baseline findings are not driven by the composition of the non-EU control group. The treated group includes 24 EU countries (excluding Denmark, Luxembourg, and Malta due to missing control data), and the control group includes 3 countries (United States, Canada, and Switzerland), for a total of 27 country clusters.

7.3 Robustness to alternative definition of bank payments

As a final robustness check, bank-based payments are redefined to include direct debits in addition to credit transfers. The outcome is therefore constructed as

$$\text{BankShare}_{ct}^{(CT+DD)} = \frac{\text{CT}_{ct} + \text{DD}_{ct}}{\text{CT}_{ct} + \text{DD}_{ct} + \text{Card}_{ct}}.$$

This definition treats both credit transfers and direct debits as bank-based instruments. Because harmonized direct-debit data are only consistently available in later years, the sample is restricted to 2022–2024, with the post-treatment indicator defined as Post = 1 for years from 2023 onward. The estimated effect is economically small and statistically indistinguishable from zero under wild cluster bootstrap inference, indicating no detectable difference between EU and control countries in this specification. Redefining bank-based payments to include direct debits therefore does not alter the baseline conclusion. Given the short sample window (2022–2024), the limited number of time periods, and the fact that PSD2 was implemented several years earlier, this exercise should be interpreted as a compositional sensitivity check rather than a fully powered causal evaluation of PSD2. Nevertheless, the result provides additional reassurance that the findings are robust to alternative constructions of payment aggregates.

Table 8: Robustness: BankShare including direct debits

EU × Post(2023+)	**0.001**
	(0.011)
Wild bootstrap p-value	0.929
Observations	69
Countries	24
Years	2022–2024

Note: Bank-based payments are defined as the sum of credit transfers and direct debits relative to total non-cash payments including credit transfers, direct debits, and card payments, i.e. $(CT + DD)/(CT + DD + Card)$. The estimation sample is restricted to 2022–2024 due to data availability. All specifications include country and year fixed effects, and standard errors are clustered at the country level. The treated group includes 24 EU countries (excluding Denmark, Luxembourg, and Malta due to missing control data), and the control group includes 2 countries (United States and Canada), for a total of 26 country clusters.

8 Discussion

The empirical results do not provide evidence that PSD2 led to an increase in bank-based account-to-account payments relative to card payments in EU countries after 2018. Across baseline difference-in-differences specifications, the EU × Post coefficient is consistently negative for both outcome measures—the credit-transfer share and the credit-transfer-to-card ratio. However, confidence intervals are wide, and wild cluster bootstrap inference indicates that these estimates are not statistically distinguishable from zero at conventional levels. The central finding is therefore the absence of robust evidence that PSD2 increased the relative use of bank-based payments at the aggregate country level. Importantly, the direction of the estimates remains stable across specifications. The coefficients remain negative when adding macroeconomic and digital infrastructure controls, when measuring payments using transaction values instead of counts, and when expanding the control group to include Switzerland. Taken together, the results do not suggest a measurable post-PSD2 shift from card-based payments toward account-to-account credit transfers in aggregate data. If anything, the point estimates lean in the opposite direction, though the lack of statistical precision warrants cautious interpretation. Several economic interpretations are consistent with this finding. First, consumer payment behavior may exhibit strong inertia. Payment habits can be highly persistent, and switching costs, including user familiarity, merchant acceptance infrastructure, and reward programs may limit rapid substitution away from established card networks. These dynamics are consistent with the economics of two-sided payment platforms, where network effects and scale economies reinforce incumbent market positions (Rochet and Tirole, 2003; Rysman, 2007). Even if PSD2 improved the technical feasibility of open banking-based payments, this may not have been sufficient to overcome entrenched two-sided market dynamics in card networks. Second, the benefits of open banking

may have accrued more strongly in financial intermediation and fintech innovation (e.g., account aggregation, lending, budgeting tools) rather than in retail payment substitution. PSD2's "access to account" (XS2A) provisions lowered barriers to entry for third-party providers, but the translation of these innovations into large-scale consumer adoption of account-to-account payments may require complementary infrastructure, merchant incentives, or pricing changes. Third, the implementation of strong customer authentication (SCA) — while designed to enhance security, it may have introduced frictions that disproportionately affected certain payment instruments during the transition phase. Transitional compliance costs and heterogeneous national enforcement may also have muted short-run competitive effects. More broadly, the post-2018 indicator likely captures a transition period, with legal implementation preceding full operational standardization and widespread consumer-facing adoption, so any effects may be gradual and difficult to detect in annual aggregate composition data. The event-study diagnostics add nuance, where post-2018 coefficients broadly align with the negative baseline estimates. On the pre-treatment side, coefficients are generally close to zero near the reference year ($k = -1$), but exhibit some non-flatness earlier in the pre-period. The placebo difference-in-differences test further supports this interpretation. By assigning the fake reform year of 2016, one can observe statistically insignificant effects, reinforcing the conclusion that the baseline results do not reflect a discrete structural break specific to PSD2 implementation. Taken together with the indexed pre-trend plots, this is best read as supportive, but not definitive evidence for parallel trends, reinforcing a cautious interpretation of the DiD estimates. The raw mean trajectories further reinforce this analysis, showing that the EU countries exhibit a smooth and persistent decline in relative credit-transfer usage throughout the entire 2012–2023 period, with no visible structural break around 2018. Moreover, individual EU country trajectories display heterogeneity in levels but not in direction, suggesting that the null result is not driven by offsetting positive or negative country-specific effects. Instead, the descriptive evidence points to a continuation of pre-existing trends rather than a reform-induced shift in payment composition. Importantly, robustness checks support the internal consistency of the findings. Including Switzerland in the control group does not materially alter the results. Using transaction values instead of counts yields directionally similar estimates, though with greater volatility. Redefining bank-based payments to include direct debits in the limited post-2022 sample does not reveal evidence of stronger substitution when the denominator is expanded to include direct debits in the late-period sample. These exercises suggest that the baseline findings are not driven by measurement choices or control-group composition. Overall, the evidence points to a gap between regulatory ambition and observable aggregate market reallocation, suggesting that PSD3 may need to focus less on expanding rules and more on implementation quality, interoperability, and the conditions needed for increased consumer and merchant

adoption.

9 Limitations and Scope Conditions

Several limitations should be considered when interpreting the results of this study.

First, the analysis relies on a cross-country difference-in-differences design with a relatively small number of country clusters. Although wild cluster bootstrap inference is employed to improve reliability in small-sample settings, statistical power remains limited. The estimates should therefore be interpreted primarily as evidence regarding the absence of large aggregate effects rather than as precise measurements of small treatment effects. Subtle shifts in payment composition may not be detectable in a country-level panel with 26 clusters. Second, the identification strategy depends on the parallel-trends assumption between EU countries and the control group (United States and Canada). Descriptive pre-trend figures and event-study diagnostics suggest broadly similar pre-2018 dynamics in indexed form, but the assumption cannot be tested directly. Structural differences between European and North American payment markets, including differences in card-network penetration, regulatory environments, and consumer payment habits, may limit the extent to which non-EU countries provide a perfect counterfactual for EU payment evolution. With only two baseline control countries, the counterfactual trend is inferred from a very limited comparison group, making the estimates more sensitive to country-specific shocks or structural changes in the control economies. Third, PSD2 implementation was neither instantaneous nor uniform across member states. Although the directive formally entered into force in 2018, key operational components, including API standardization, third-party-provider licensing, and Strong Customer Authentication (SCA) enforcement where all introduced gradually and with heterogeneous timing across countries. As a result, the "Post-2018" indicator may capture a transition period rather than a fully operational open-banking regime. This staggered implementation could attenuate estimated treatment effects in aggregate panel data. Fourth, the outcome variables measure aggregate payment composition rather than direct adoption of PSD2-enabled services. Credit transfers include a broad set of instruments beyond open-banking payment initiation services, such as traditional bank transfers and instant payments unrelated to XS2A usage. The analysis therefore evaluates whether PSD2 altered the overall balance between bank-based and card-based payments, not whether open-banking applications themselves expanded. If PSD2 primarily affected niche segments of the payments market, such effects may not be detectable in country-level aggregates. Fifth, measurement constraints affect the treatment of payment instruments. Direct debits are excluded from baseline outcomes due to incomplete early coverage, and value-based measures may reflect transaction size rather than frequency of use. While robustness exercises yield qualitatively

similar conclusions, the available data limit the ability to analyse the full composition of bank-based payments consistently over the entire sample period. Finally, the time horizon of the study may be insufficient to capture long-run effects of open-banking regulation. Payment systems are characterised by strong network effects, infrastructure dependence, and persistent consumer habits. Regulatory changes that lower entry barriers may require many years before translating into measurable shifts in aggregate payment behaviour. The results should therefore be interpreted as evidence about the medium-run impact of PSD2 (2018–2023), rather than its long-run consequences. Taken together, these limitations imply that the findings should be understood as evidence regarding whether PSD2 produced observable aggregate substitution from card payments toward bank-based payments during the first years after implementation, rather than as a comprehensive evaluation of open banking as a technological or regulatory innovation.

10 Conclusion

This thesis examined whether the introduction of PSD2 was associated with a measurable shift in payment composition from card-based payments toward bank-based account-to-account transfers in EU member states. Using harmonised country–year panel data and a difference-in-differences framework comparing EU countries to non-EU controls before and after 2018, the analysis finds no evidence of a positive reallocation toward credit transfers following PSD2 implementation. Point estimates are consistently negative across baseline specifications and robustness checks, though statistically imprecise under wild cluster bootstrap inference. These findings carry important implications for digital competition policy. If PSD2 had produced a clear and statistically robust shift from credit cards toward bank-based payments, it would have strengthened the case for open banking as an effective regulatory model capable of reshaping entrenched network markets. Instead, the results suggest that regulatory access provisions alone may be insufficient to displace established payment ecosystems in the medium run. At the level of aggregate country-year payment composition, PSD2 does not appear to have materially altered the relative dominance of card-based payments during the period studied. The absence of a detectable aggregate substitution effect is reinforced by dynamic event-study diagnostics, raw mean comparisons, country-level pre–post change calculations and placebo test that fails to detect spurious treatment effects under alternative reform timing. Descriptive evidence shows that most EU member states experienced declines in the relative use of bank-based payments between the pre- and post-period averages, suggesting that the aggregate null result does not mask a subset of countries with strong positive effects. Expanding the control group to include Switzerland, using transaction values instead of counts, and redefining bank-based payments to incorporate direct debits in a

restricted late-period sample all yield qualitatively similar conclusions. In sum, these exercises indicate that the central finding is not driven by control-group composition, outcome definition, or measurement choice. This does not imply that PSD2 failed in broader terms. The directive may have improved consumer data portability, stimulated fintech innovation, and enhanced competitive entry in ways not immediately reflected in aggregate payment shares. Open banking may operate through gradual ecosystem changes rather than rapid substitution in payment composition. However, the absence of measurable substitution in core payment aggregates highlights the difficulty of translating regulatory design into observable market-level behavioral change within a relatively short post-implementation window. As the European Union considers the next iteration of the payments framework under PSD3, these findings underscore a key lesson: the challenge may not lie primarily in expanding regulatory mandates, but in ensuring that existing open banking provisions translate into tangible consumer-facing adoption and competitive outcomes. Strengthening implementation consistency, interoperability standards, merchant incentives, and user experience may prove more consequential than broadening formal access rules alone. Future research using more granular transaction-level data, institution-level heterogeneity, or consumer-level adoption measures would be valuable for identifying the specific channels through which open banking can influence market structure over longer time periods. Nevertheless, within the scope of aggregate cross-country evidence between 2012–2023, this thesis provides a disciplined empirical assessment of PSD2's observable impact on payment composition and clarifies the medium-run boundaries of regulatory effectiveness in digital financial markets.

A Appendix

A.1 Additional Pre-treatment Trend Diagnostics

This appendix reports supplementary pre-treatment trend diagnostics for the alternative baseline outcome $\text{BankCardRatio}_{ct} = CT/Card$, which measures the relative intensity of bank-based credit transfers compared to card payments. These figures complement the main-text analysis based on BankShare_{ct} and provide additional reassurance regarding the plausibility of the parallel trends assumption. Figures 9 and 10 plot average trends for EU member states and for the control group (United States and Canada) over the pre-PSD2 period 2012–2017. As in the main analysis, outcomes are constructed using transaction counts only, and the EU sample is restricted to countries with complete baseline coverage. In levels, the $\text{BankCardRatio}_{ct}$ outcome displays smooth and broadly similar downward trends across the EU and the control group. When indexed to a common baseline year (2012 = 100), the growth paths are closely aligned, with no evidence of anticipatory divergence prior to PSD2 implementation. These patterns support the conclusion drawn from the main-text figures that pre-treatment dynamics are sufficiently similar to justify the difference-in-differences design.

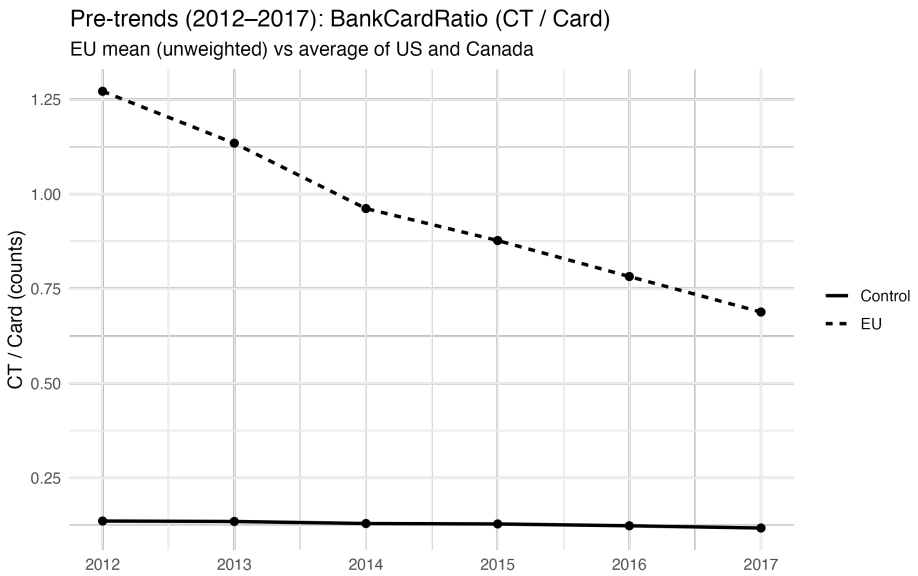


Figure 9: Pre-treatment trends in levels (2012–2017): $\text{BankCardRatio}_{ct} = CT/Card$. EU mean (unweighted) versus the average of the United States and Canada.
Note: The treated group includes 24 EU countries (excluding Denmark, Luxembourg, and Malta due to missing control data), and the control group includes 2 countries (United States and Canada), for a total of 26 country clusters.

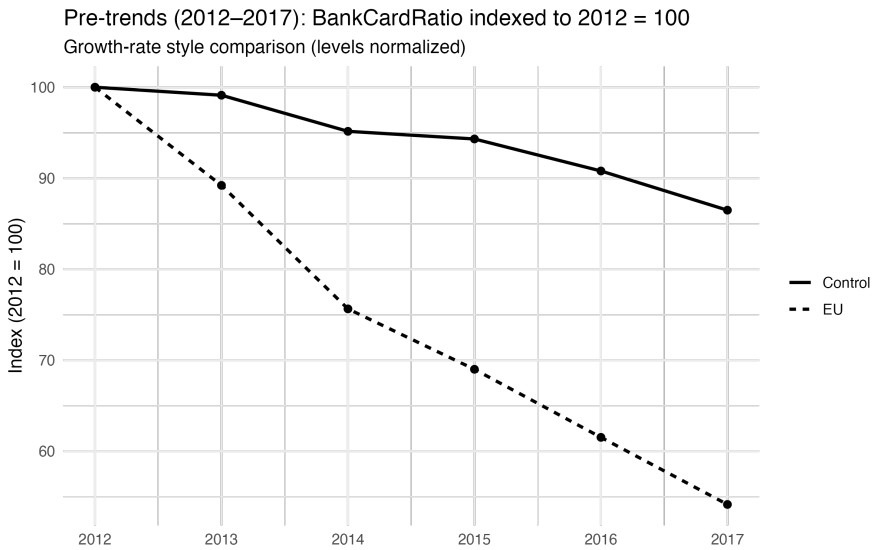


Figure 10: Pre-treatment trends indexed to 2012 = 100 (2012–2017): BankCardRatio$_{ct}$ = $CT/Card$. EU mean (unweighted) versus the average of the United States and Canada. *Note:*The treated group includes 24 EU countries (excluding Denmark, Luxembourg, and Malta due to missing control data), and the control group includes 2 countries (United States and Canada), for a total of 26 country clusters.

A.2 Country-level raw changes before vs after PSD2

To assess whether the aggregate null or negative result masks heterogeneity within the treated group, I compute for each EU country the change in average outcomes between the pre period (2012–2017) and the post period (2018–2023):

$$\Delta y_c = \overline{y}_{c,2018-2023} - \overline{y}_{c,2012-2017}.$$

Figures 11 and 12 plot these country-level changes for the EU sample, with the corresponding changes for the United States and Canada shown as reference values.

The dispersion in country-level changes provides a simple diagnostic of whether some EU members exhibit marked post-reform shifts even if the EU mean does not. The results show that the majority of EU countries experienced a decline in the relative use of bank-based payments between the pre- and post-period averages. Only a small number of countries display modest positive changes, and these are generally small in magnitude. Importantly, the pattern of raw changes is broadly consistent with the baseline difference-in-differences estimates: there is no clear cluster of EU countries exhibiting large positive shifts after 2018 relative to their own pre-period averages. This descriptive evidence reinforces the interpretation that the aggregate null finding is not driven by offsetting positive and negative country-specific effects. This exercise is purely descriptive (no fixed effects or controls) and is included to contextualize the regression results rather than to form a separate causal claim.

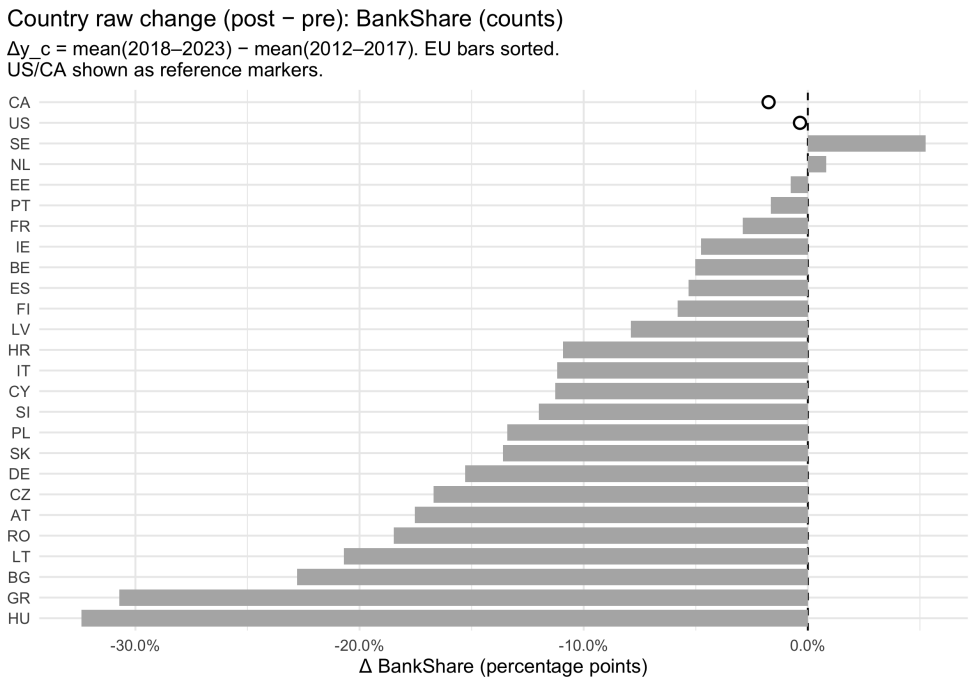


Figure 11: Country-level raw change in BankShare (counts): average 2018–2023 minus average 2012–2017. EU countries shown individually; US/CA shown as references.
Note: The treated group includes 24 EU countries (excluding Denmark, Luxembourg, and Malta due to missing control data), and the control group includes 2 countries (United States and Canada), for a total of 26 country clusters.

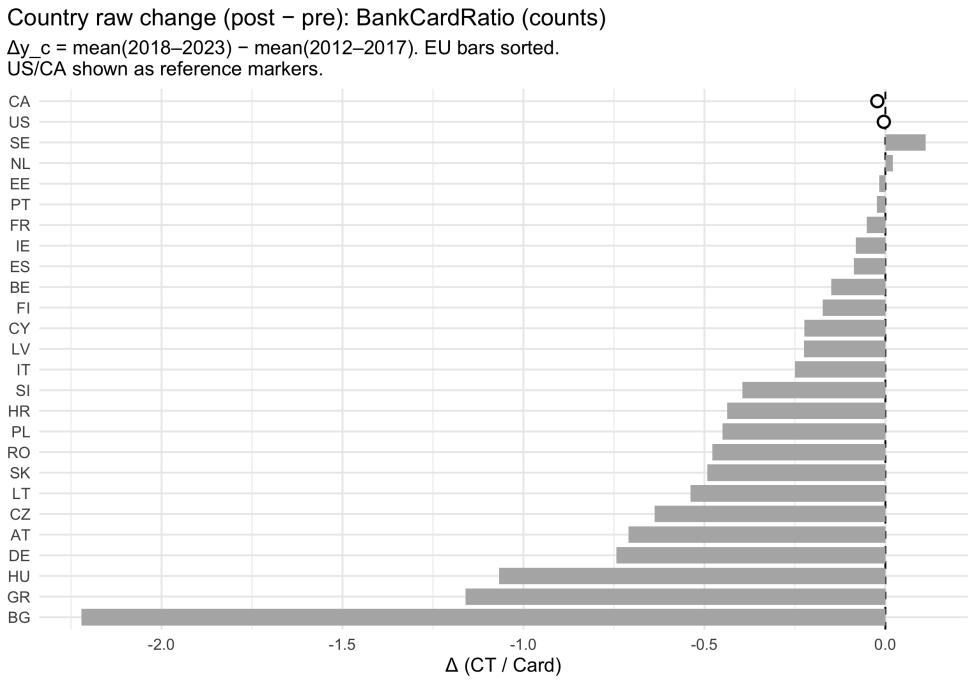


Figure 12: Country-level raw change in BankCardRatio (counts): average 2018–2023 minus average 2012–2017. EU countries shown individually; US/CA shown as references.
Note: The treated group includes 24 EU countries (excluding Denmark, Luxembourg, and Malta due to missing control data), and the control group includes 2 countries (United States and Canada), for a total of 26 country clusters.

A.3 EU country trajectories over time

Figure 13 plots the evolution of BankShare (counts) for each EU member state over the full sample period (2012–2023). Thin lines represent individual countries, while the thick line shows the unweighted EU mean. The vertical dashed line marks the implementation year of PSD2 (2018). The figure highlights substantial cross-country heterogeneity in levels of bank-based payment usage. However, a common pattern emerges: most countries exhibit a gradual decline in BankShare during the pre-2018 period, and this downward trajectory generally continues after 2018. There is no clear cluster of countries displaying a marked upward shift following PSD2 implementation. This visual evidence suggests that the aggregate null result is unlikely to be driven by offsetting positive and negative country-specific effects. Rather, the absence of a detectable treatment effect at the EU level reflects a broadly shared pattern across member states.

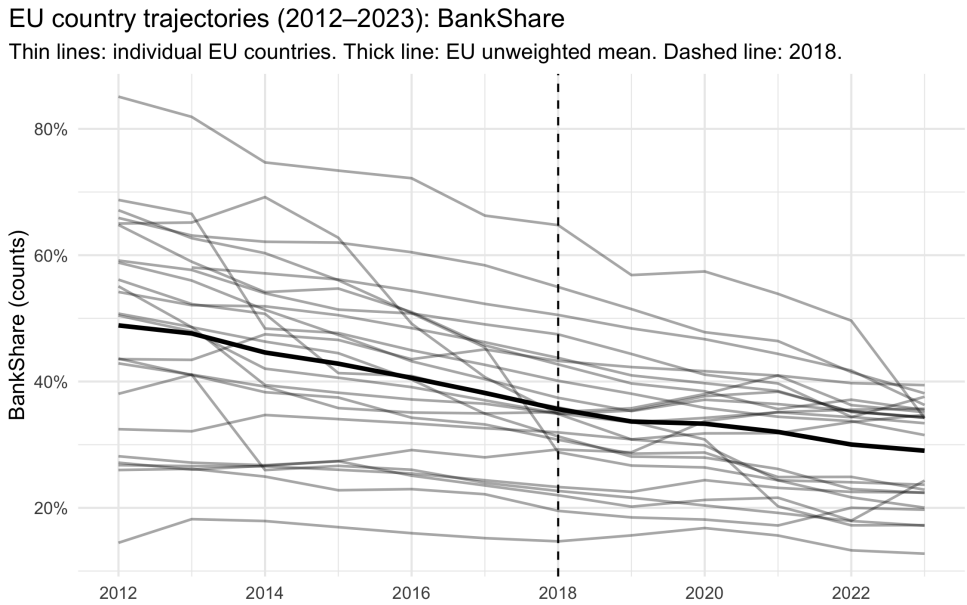


Figure 13: EU country trajectories (2012–2023): BankShare (counts). Thin lines show individual EU member states; the thick line shows the unweighted EU mean. The dashed vertical line marks 2018 (PSD2 implementation).
Note: The treated group includes 24 EU countries (excluding Denmark, Luxembourg, and Malta due to missing control data), and the control group includes 2 countries (United States and Canada), for a total of 26 country clusters.

A.4 Event-study results and diagnostic: BankCardRatio (counts-based)

This appendix reports the corresponding event-study coefficients for the BankCardRatio outcome (constructed from counts only). The specification, fixed effects structure, and inference approach match the main event-study diagnostic in Section 6.5. Coeffi-

cients are reported relative to 2017 ($k = -1$), with event time k = year − 2018.

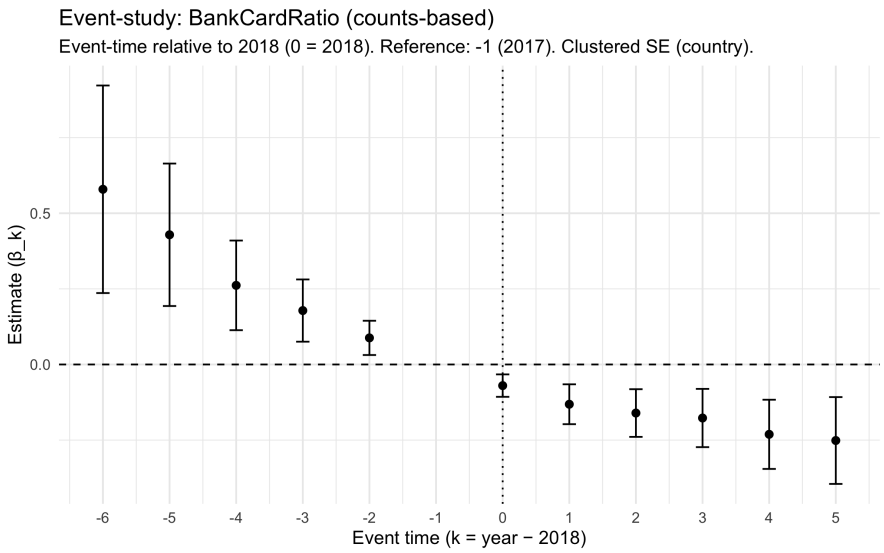


Figure 14: Event-study estimates for BankCardRatio (counts-based). Coefficients are relative to 2017 ($k = -1$); $k = 0$ corresponds to 2018. Country and year fixed effects included. Standard errors clustered at the country level.
*Note:*The treated group includes 24 EU countries (excluding Denmark, Luxembourg, and Malta due to missing control data), and the control group includes 2 countries (United States and Canada), for a total of 26 country clusters.

A.5 Values-based estimates without controls

Table 9 reports the baseline estimates. The point estimates are negative for both outcomes, indicating a relative decline in bank-based payment values in EU countries after the implementation of PSD2. While the estimated effects are not statistically significant under wild bootstrap inference, their sign and magnitude are consistent with the corresponding counts-based results. The lack of statistical precision for value-based outcomes is not unexpected. Payment values are substantially more volatile than transaction counts and are sensitive to cross-country price levels, large-ticket transactions, and institutional reporting differences. For this reason, values-based estimates are interpreted as complementary evidence rather than the primary estimand.

Table 9: Baseline difference-in-differences estimates using transaction values

	BankShare (values)	BankCardRatio (values)
EU × Post(2018+)	**-0.017**	**-39.502**
	(0.007)	(10.643)
Wild bootstrap p-value	0.174	0.155
Observations	305	305
Countries	26	26
Years	2012–2023	2012–2023

*Note:*The treated group includes 24 EU countries (excluding Denmark, Luxembourg, and Malta due to missing control data), and the control group includes 2 countries (United States and Canada), for a total of 26 country clusters.

A.6 Expanded control group: inclusion of Switzerland

As an additional robustness check, Switzerland is added to the baseline control group alongside the United States and Canada. This expansion increases the number of control clusters and provides a broader counterfactual benchmark for EU payment trends. Figures 15 and 16 report the baseline and control-augmented estimates under this alternative comparison group. The estimated EU × Post coefficients remain negative and similar in magnitude to the baseline results. Confidence intervals overlap closely with the original specifications, and no statistically significant treatment effect emerges under wild cluster bootstrap inference. These results indicate that the baseline findings are not driven by reliance on a two-country control group and are robust to modest expansion of the counterfactual comparison set.

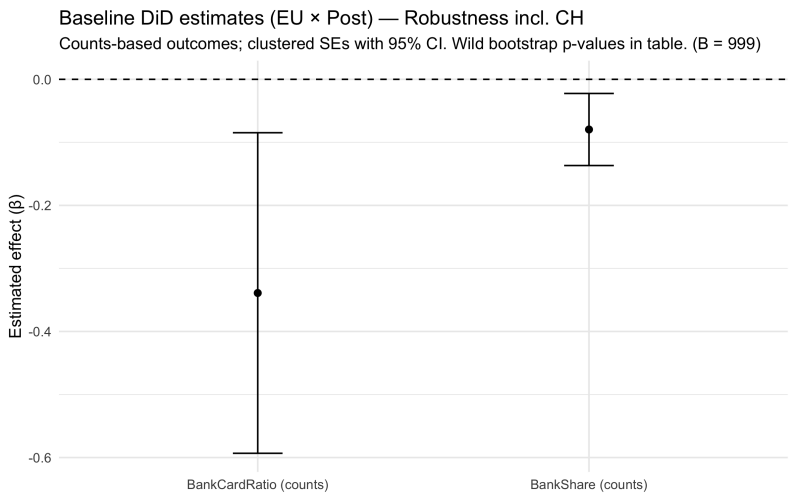


Figure 15: Baseline DiD estimates (EU × Post) with expanded control group including Switzerland. Error bars indicate 95% confidence intervals based on country-clustered standard errors. Wild cluster bootstrap p-values are reported in the corresponding table (B = 999).

*Note:*The treated group includes 24 EU countries (excluding Denmark, Luxembourg, and Malta due to missing control data), and the control group includes 3 countries (United States, Canada and Switzerland), for a total of 27 country clusters.

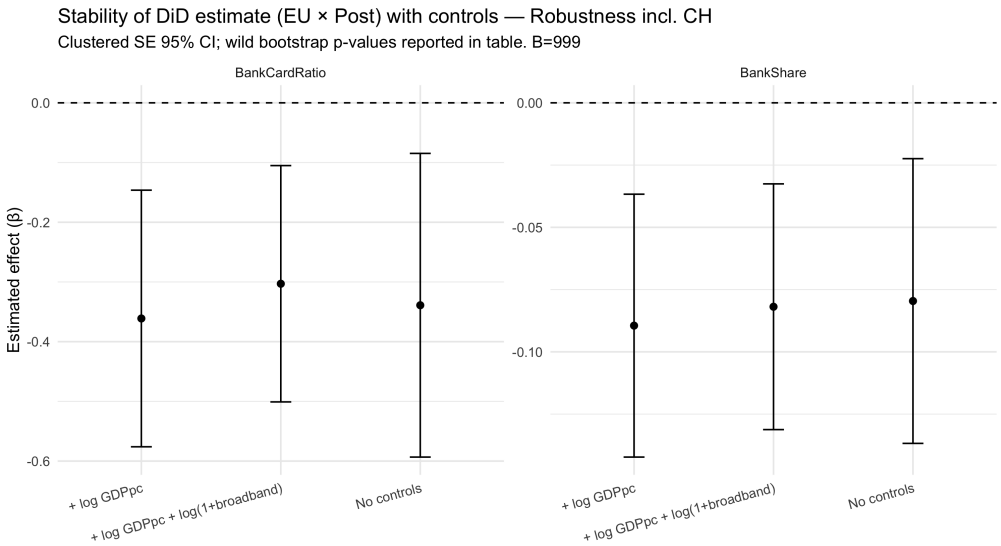


Figure 16: Stability of DiD estimates with controls, including Switzerland

Note: The figure plots coefficient estimates and 95% confidence intervals for the EU × Post coefficient across specifications when Switzerland is added to the control group. Estimates are obtained from country–year panel regressions with country and year fixed effects. Confidence intervals are based on standard errors clustered at the country level, and wild cluster bootstrap p-values are reported in the corresponding table.
The pattern closely mirrors the baseline stability results: point estimates are consistently negative for both BankShare and BankCardRatio and remain within overlapping confidence intervals across specifications. This indicates that the estimated PSD2 association is stable to both the inclusion of additional controls and the expansion of the control group. The treated group includes 24 EU countries (excluding Denmark, Luxembourg, and Malta due to missing control data), and the control group includes 3 countries (United States, Canada and Switzerland), for a total of 27 country clusters.

A.7 Alternative definition of bank-based payments

This appendix presents the graphical analogue of the robustness exercise in which bank-based payments are redefined to include direct debits alongside credit transfers. The outcome is constructed as $(CT + DD)/(CT + DD + Card)$ using the restricted sample 2022–2024. Consistent with the main-text results, the estimated EU × Post coefficient is economically small and statistically indistinguishable from zero. The confidence interval spans zero comfortably, indicating no detectable post-PSD2 shift toward bank-based instruments under this alternative aggregate definition.

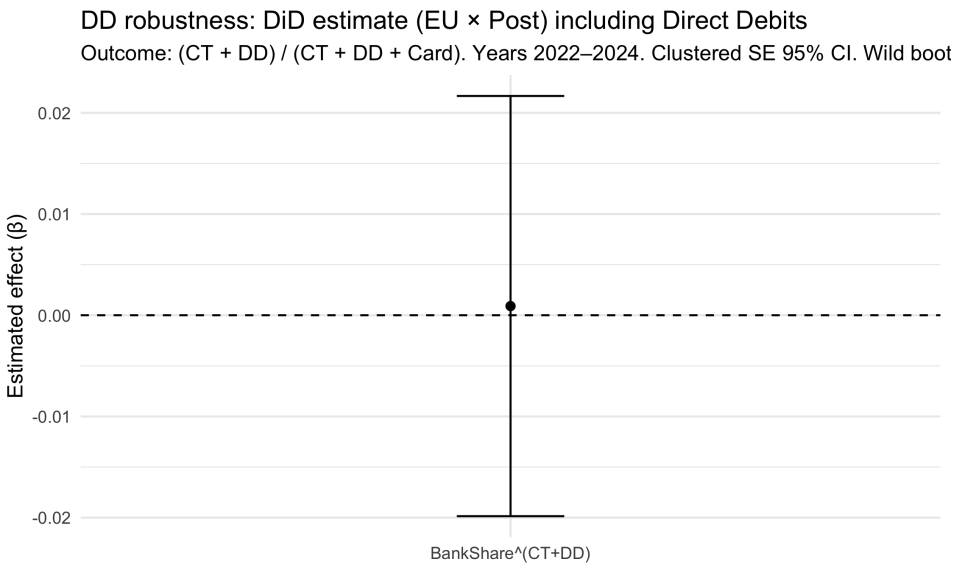


Figure 17: Robustness to including direct debits in the bank-based aggregate. DiD estimate (EU × Post) for $\text{BankShare}_{ct}^{(CT+DD)} = (CT + DD)/(CT + DD + Card)$ using the restricted sample 2022–2024 and defining Post = 1 for years ≥ 2023. Error bars show 95% confidence intervals based on country-clustered standard errors; wild cluster bootstrap p-values are reported in Table 8.

Note: The treated group includes 24 EU countries (excluding Denmark, Luxembourg, and Malta due to missing control data), and the control group includes 2 countries (United States and Canada), for a total of 26 country clusters.

A.8 Conceptual heterogeneity extension.

To explore whether the treatment effect varies with pre-existing payment structure, countries can be classified by baseline card exposure prior to PSD2. Let HighExposure_c be an indicator equal to one for countries with above-median card usage in 2015. A potential triple-difference specification would take the form:

$$y_{ct} = \alpha_c + \lambda_t + \beta_1(\text{EU}_c \times \text{Post}_t) + \beta_2(\text{EU}_c \times \text{Post}_t \times \text{HighExposure}_c) + \varepsilon_{ct}.$$

The coefficient β_2 would capture whether the PSD2 treatment effect differs between high-card-exposure and low-card-exposure countries. Because this extension adds an additional layer of identifying assumptions and reduces transparency relative to the baseline design, it is treated as an exploratory extension and left for future work. Nevertheless, the construction of a baseline exposure measure illustrates how heterogeneous treatment effects could be evaluated in a natural extension of the baseline difference-in-differences design. In particular, countries with higher pre-PSD2 card usage may exhibit smaller substitution toward bank-based payments if card adoption was already advanced prior to the regulatory change.

Table 10: Baseline card exposure by country (2015)

Country	Card exposure (2015)	Exposure group
PT	0.8305381238583466	High card exposure
ES	0.7720975229497464	High card exposure
FR	0.7396481858158905	High card exposure
CY	0.7334878331402085	High card exposure
IE	0.7260528571732205	High card exposure
SE	0.7259504975759122	High card exposure
EE	0.6594712006591267	High card exposure
NL	0.6420067282459498	High card exposure
IT	0.6253118252968384	High card exposure
FI	0.6176876057383315	High card exposure
LV	0.5939798168226259	High card exposure
LT	0.5866266131361235	High card exposure
RO	0.5550795765548128	Low card exposure
BE	0.5341509698243491	Low card exposure
PL	0.5262162282870955	Low card exposure
AT	0.5231862327821764	Low card exposure
SI	0.49492445900370613	Low card exposure
SK	0.4861079617452081	Low card exposure
CZ	0.45278360479794133	Low card exposure
HU	0.4396108051960798	Low card exposure
HR	0.4383986806996739	Low card exposure
DE	0.3800690996154713	Low card exposure
GR	0.3723526878614953	Low card exposure
BG	0.2660760398606281	Low card exposure

Note: Baseline card exposure is measured in 2015 as the share of card payments in total non-cash transactions (card plus credit transfers). Countries are classified into high- and low-exposure groups based on a median split. This classification is constructed solely to illustrate a potential heterogeneity extension of the baseline difference-in-differences design and is not used for estimation in the present analysis.

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